

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import re
import nltk
nltk.download('stopwords')
from nltk.corpus import stopwords
import joblib
```

```
stop_words = set(stopwords.words('english'))
```

[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.

```
import joblib
import numpy as np
import nltk
nltk.download('stopwords', quiet=True)
from nltk.corpus import stopwords
from sklearn.utils.validation import check_is_fitted
from tensorflow import keras
```

```
stop_words = set(stopwords.words('english'))
```

```
# — Load vectorizer —————
try:
    vectorizer = joblib.load('/content/drive/MyDrive/sentiment_vectorizer.joblib')
    print("Vectorizer loaded )
except Exception as e:
    print(f"Error loading vectorizer: {e}")
```

```
# — Load Logistic Regression —————
try:
    log_reg_model = joblib.load('/content/drive/MyDrive/sentiment_logistic_regression.joblib')
    check_is_fitted(log_reg_model)
    print("Logistic Regression loaded )
except Exception as e:
    print(f"Error loading Logistic Regression: {e}")
```

```
# — Load Naive Bayes —————
try:
    naive_model = joblib.load('/content/drive/MyDrive/sentiment_naive_bayes.joblib')
```

```

bayes_model = LogisticRegression()
check_is_fitted(bayes_model)
print("Naive Bayes loaded ✅")
except Exception as e:
    print(f"Error loading Naive Bayes: {e}")

# — Load ANN —————
try:
    ann_model = keras.models.load_model('/content/drive/MyDrive')
    print("ANN loaded ✅")
except Exception as e:
    print(f"Error loading ANN: {e}")

```

Vectorizer loaded ✅

Logistic Regression loaded ✅

Naive Bayes loaded ✅

WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty

ANN loaded ✅

```

def clean_tweet(text):
    text = text.lower()
    text = re.sub(r'http\S+|www\S+', '', text)
    text = re.sub(r'@\w+', '', text)
    text = re.sub(r'#\w+', '', text)
    text = re.sub(r'^a-z\s]', '', text)
    words = text.split()
    words = [w for w in words if w not in stop_words]
    return ' '.join(words)

```

```

def predict_sentiment(text):
    # Clean and vectorize
    cleaned = clean_tweet(text)
    vectorized = vectorizer.transform([cleaned])

    # — Logistic Regression —————
    lr_prob = max(log_reg_model.predict_proba(vectorized)[0])
    lr_label = log_reg_model.predict(vectorized)[0]
    lr_sentiment = "Positive 😊" if lr_label == 1 else "Negative 😞"

    # — Naive Bayes —————
    nb_prob = max(bayes_model.predict_proba(vectorized)[0])
    nb_label = bayes_model.predict(vectorized)[0]
    nb_sentiment = "Positive 😊" if nb_label == 1 else "Negative 😞"

    # — ANN —————
    ann_prob = ann_model.predict(vectorized to array())
    ann_label = ann_model.predict(vectorized to array())
    ann_sentiment = "Positive 😊" if ann_label == 1 else "Negative 😞"

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ann_prob = ann_model.predict(vectorized.texts[0], verbose=0)
ann_label = 1 if ann_prob > 0.5 else 0
ann_sentiment = "Positive 😊" if ann_label == 1 else "Negative 😞"
ann_confidence = ann_prob if ann_label == 1 else 1 - ann_prob

# — Output —
print(f"\nInput: \"{text}\"")
print(f"\nLogistic Regression → {lr_sentiment} ({round(lr_confidence, 2)}%)")
print(f"Naive Bayes → {nb_sentiment} ({round(nb_confidence, 2)}%)")
print(f"ANN → {ann_sentiment} ({round(ann_confidence, 2)}%)")

```

```

user_input = input("Enter a sentence: ")
predict_sentiment(user_input)

```

Enter a sentence: I used to love this.

Input: "I used to love this. "

Logistic Regression	→	Positive 😊	(81.3%)
Naive Bayes	→	Positive 😊	(64.5%)
ANN	→	Negative 😞	(60.9%)

try these: -The food was amazing but the service was terrible.

-Wow, my flight got cancelled. Best day ever.

-bro!!! the missiles will destroy the world #saveEarth

-i used to love this

-Nepal

✓ Improved ANN (100k data + negation fix)

```

import joblib
from tensorflow import keras

# Load improved vectorizer and model
vectorizer_improved = joblib.load('/content/drive/MyDrive/sentiment_vectorizer.pkl')
ann_improved = keras.models.load_model('/content/drive/MyDrive/ann_model.h5')

print("Improved ANN loaded! ✅")

```

WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty.
Improved ANN loaded! ✅

```

def predict_sentiment_improved(text):
    cleaned = clean_tweet(text)
    vectorized = vectorizer_improved.transform([cleaned]).toarray()

    prob = ann_improved.predict(vectorized, verbose=0)[0][0]
    label = 1 if prob > 0.5 else 0

```

```
sentiment = "Positive 😊" if label == 1 else "Negative"
confidence = round(float(prob if label == 1 else 1 - prob), 1)

print(f"\nInput: \"{text}\"")
print(f"Improved ANN → {sentiment} ({confidence}%")
```

```
user_input = input("Enter a sentence: ")
predict_sentiment_improved(user_input)
```

Enter a sentence: i cant believe what just happened

Input: "i cant believe what just happened"
Improved ANN → Negative 😞 (96.1%)